Satyam Patel

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EDUCATION

Stevens Institute of Technology Hoboken, NJ
Bachelor of Engineering in Mechanical Engineering, Minor in Systems and Industrial Engineering. GPA: 3.2 / 4.0. *Expected May 2029*

SKILLS

Manufacturing & Machining: Precision Machining, Part Inspection, Quality Control, Manual Lathes, Milling, Welding, 3D Printing (Bambu Lab P2S), Cura Slicer
Design & Simulation: SolidWorks 2025, Onshape, Canva
Software & Systems: Arduino IDE, C++, MQTT, Google Workspace, Microsoft 365, Cloud Data Streaming
Soft Skills: Project Management, Cross-functional Team Leadership, Technical Communication

EXPERIENCE

Intern, Precision Machining & Quality Control May 2026 – August 2026
Precision Forms *Butler, NJ*

- Performed precision machining, part inspection, and quality control on high-precision mechanical components using micrometers, maintaining strict tolerances as low as five tenths of an inch.
- Supported manufacturing processes for aerospace and medical-grade components, working within strict industry standards.
- Completed a capstone project involving the custom machining of a metal barrel for a Pentel P207 mechanical pencil, applying tight-tolerance turning and assembly design.

PROJECTS

Autonomous Rover
Lead Mechanical Designer & Systems Programmer

- Engineered a closed-loop autonomous rover using a WeMos D1 R2 (ESP8266), Pololu Zumo chassis, and a 3-sensor ultrasonic array for dynamic obstacle detection and wall tracking.
- Designed an open-frame chassis attachment, snap-fit OLED mount, and ArUco marker platform in SolidWorks, reducing print time by over 50% to 1 hr 30 min.
- Programmed multi-threaded C++ firmware on a 100 ms execution loop, integrating real-time MQTT coordinates with proportional angle-steering and ultrasonic avoidance logic.
- Navigated a 4-waypoint obstacle arena in 32 seconds with zero collisions during final validation testing, against a course requirement of under 3 minutes.

Smart Autonomous Watering System
Electrical & Product Designer

- Developed a self-contained IoT device to monitor soil moisture, light, and humidity using Arduino microcontrollers.
- Designed a custom modular housing in SolidWorks 2025 to integrate all mechanical and electronic components.
- Programmed logic for real-time data streaming to a cloud server while simultaneously managing pump actuation.

ACTIVITIES & LEADERSHIP

Boy Scouts of America (Patriots Path Council) June 2018 – Present
Eagle Scout

- Achieved Eagle Scout rank (attained by only 6% of Scouts), completing a comprehensive community service capstone project. Inducted into the Order of the Arrow, BSA's National Honor Society, for exemplary service and leadership.

Vex Robotics Team Sept 2023 – June 2025
Lead Engineer & Captain

- Led design and assembly of competitive robots, ensuring compliance with all Vex technical regulations.
- Mentored teammates on mechanical assembly and iterative design processes to address field challenges.

Drone Racing Club Sept 2020 – June 2025
Vice President

- Managed equipment inventory and performed technical troubleshooting and repairs on racing drones.
- Facilitated training sessions for new members on flight mechanics and safety protocols.

AWARDS & HONORS

- Eagle Scout Rank (2025)
- National Youth Leadership Training (NYLT) Graduate
- Sea Base Captains Club Award: Recognized for leadership and teamwork in high-adventure marine environments